

CAPSTONE CURRENT SYSTEM GAP ANALYSIS REPORT

A Gamified Simulation Platform with Automated Skill Evaluation
for NC II Computer Systems Servicing

Report type	Current-state product, architecture, security, QA and requirements compliance audit
Audit date	07 August 2026
Audit mode	Strictly read-only
Systems	Flutter learner application; Next.js web dashboard; referenced shared data services
Primary source	Capstone_PRD_Gamified_CSS_NCII.pdf, Version 1.0, 25 pages
Prepared for	Development reference, capstone documentation, adviser review and final-defense preparation

No application code, configuration, dependency, database record, migration, user interface, commit or remote repository state was changed during this audit. Only temporary audit-rendering files and this report artifact were created.

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The report follows the required audit structure. Major evidence matrices and the detailed gap register may span multiple printed pages.

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Status vocabulary: COMPLETE, PARTIAL, MISSING, INCORRECT, UNVERIFIED and OUT OF SCOPE.

Document Control

Control	Value
Project	A Gamified Simulation Platform with Automated Skill Evaluation for NC II Computer Systems Servicing
Report	Current System Gap Analysis and Development Readiness Audit
Version / date	Audit Report 1.0 / 07 August 2026 (Asia/Manila)
Audit mode	Read-only static inspection; no implementation or remediation
Systems reviewed	Flutter/Dart learner app; Next.js/React/TypeScript dashboard; Firebase client code; Supabase mobile integration; local persistence; tests; configuration references; documentation and generated-artifact boundaries
Source-of-truth order	PRD PDF, study objectives, approved scope/limitations, panel recommendations, existing implementation
Repository state note	No .git metadata was present in the supplied workspace, so a Git diff could not be used as an audit control. No commit or push command was issued.

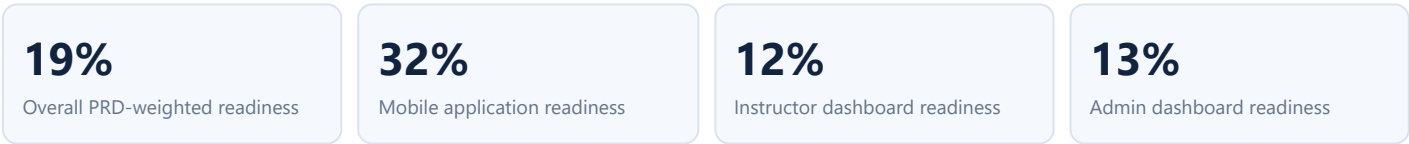
Status Legend

COMPLETE	Strong static evidence satisfies the requirement.	PARTIAL	Meaningful implementation exists, but material capability is absent.
MISSING	No meaningful implementation found.	INCORRECT	Implementation exists but conflicts with or cannot satisfy the PRD.
UNVERIFIED	Code/reference exists, but runtime or external configuration is unavailable.	OUT OF SCOPE	Not required by the approved baseline.

Priority: **P0** objective-critical or mandatory; **P1** important compliance, integrity or workflow requirement; **P2** enhancement or polish.

Executive Summary

The current repository is a useful demonstration foundation, but it cannot yet demonstrably satisfy the approved PRD or the five study objectives as an integrated system. The learner app contains substantive interactive work; the web application is primarily a UI prototype; and the required shared, secured, versioned evaluation platform is absent from the supplied codebase.



Acceptance-blocking conclusion. Objective 4 is not demonstrable: scores are calculated client-side using hard-coded percentages and thresholds, no approved TESDA source/rubric version is represented, criterion-level evaluation is absent, and no instructor finalization/release workflow or immutable score revision exists.

What exists now • Twenty mobile missions across four locally named COCs. • Text scenario briefs and five interaction patterns: identification, drag-and-drop, configuration, procedure selection and troubleshooting. • Mobile XP, points, levels, badges/achievements, leaderboard and progress code using Supabase plus SharedPreferences. • Responsive web dashboard shells for modules, users, progress, analytics, reports, settings and logs. • Client-side CSV export and generic Firebase Firestore helper functions.	What prevents acceptance • Web authentication bypass; any syntactically valid email/password is accepted. • Instructor and Admin merged into <code>instructor_admin</code>; no separate workspace or authorization. • No classes, enrollments, assignments, learning-resource files, quizzes, COC bypass, score review, finalization or release. • Web dashboards and reports use dummy data; mobile and web use different providers and records. • No server API, repository database schema/migrations, Firebase rules, Supabase policies, idempotency or versioned rubric. • No meaningful tests for objective-critical behavior.
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Principal Audit Judgment

The mobile implementation supports a credible interactive prototype for Objective 1 and portions of Objective 2. It does not establish TESDA traceability, authoritative automated evaluation, or an auditable instructor-reviewed assessment lifecycle. The web implementation presents intended screens but does not provide trustworthy operational evidence for Objectives 4 or 5. Development should begin with identity, shared data ownership, versioned criteria and submission integrity - not additional cosmetic screens.

Audit Scope

The audit covered the entire supplied PRD, the production-source and configuration surfaces of both applications, referenced persistence models, route and permission boundaries, scoring/evaluation logic, gamification, analytics, tests, platform manifests and supporting specifications.

Included	Excluded from implementation credit
All 25 PRD pages; 83 Dart files under lib/; all mobile tests and manifests; 107 web TS/TSX/CSS source files; routes, hooks, services, types and configuration references.	node_modules, .next, Flutter build/generated/ephemeral output and prose implementation summaries. Generated artifacts were inventoried but not treated as authoritative source.
Static security review of authentication, authorization, direct data access, credential handling, validation, upload surfaces and auditability.	Penetration testing, destructive security validation, live-account manipulation, network interception or database mutation.
Static UI/UX and accessibility evidence: layouts, states, controls, asset use and responsiveness code.	Redesign, visual changes, device-lab testing or user research.

Runtime limitation. Tests/builds were not executed because the supplied mobile database test creates users and inserts live Supabase records, while the audit mandate forbids database changes. Type-check/build commands can also write cache artifacts. Runtime-only behavior is therefore labeled UNVERIFIED.

Audit Methodology

1. Extracted and read all 25 PRD pages, then rendered every page and visually reviewed five contact sheets for completeness.
2. Built repository inventories excluding vendor/build output; identified entry points, routes, state, services, persistence calls, models, assets and tests.
3. Traced each objective and requirement from UI to state/service, data write/read and authorization boundary where code existed.
4. Distinguished UI presence, hard-coded demonstration data, client-side persistence and verified backend enforcement.
5. Applied conservative statuses; COMPLETE required direct code evidence across the necessary layers.
6. Scored readiness using priority and status weights documented in Section 45.

Evidence Rules

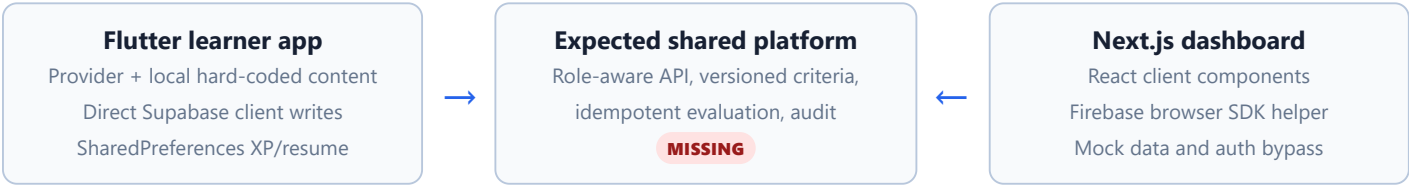
Paths, component/service names and endpoints are reported only when present. Missing database definitions and external RLS/rules are not inferred from client calls. Planning documents were used to identify acknowledged gaps but never to upgrade implementation status.

Source-of-Truth Hierarchy

Rank	Source	Audit application
1	PRD PDF	Controls product, role, evaluation, data, UI and quality requirements.
2	Objectives of the Study	Determines whether the implemented product can support the research claims.
3	Approved Scope and Limitations	Prevents ERP/SIS/enterprise-LMS expansion and prohibits official TESDA claims.
4	Panel Recommendations	Provides the 14 mandatory/optional correction themes.
5	Existing code	Evidence of current state; it does not redefine a conflicting requirement.

No-guess rule retained. TESDA value/rule requires validation against the approved TESDA CSS NC II booklet. No percentage, pass threshold, rubric, weight or competency value in this report is endorsed as TESDA-approved.

Current System Architecture Overview



Layer	Current evidence	Architectural consequence
Mobile UI/domain	ByteQuest Mobile App/lib; 20 locally defined missions; Provider services.	Substantial learner prototype, but content and rules are duplicated in code.
Mobile persistence	SupabaseConfig; 16 referenced tables/views; SharedPreferences gamification/resume.	Split local/remote truth and direct client trust; schema and RLS unavailable.
Web UI/domain	src/app; client pages, hard-coded arrays and redirects.	Screen presence does not prove business workflow or persistence.
Web persistence	src/services/firebase.service.ts generic browser CRUD.	No server validation or scoped authorization in repository.
Identity	Mobile Supabase Auth; web disabled Firebase Auth and localStorage flag.	No common identity, role or account-state authority.
Backend/API	No src/app/api, middleware, server controllers, schema, migrations or rules files.	PRD shared-service and security requirements cannot be demonstrated.

Repository / Codebase Inventory

25

PRD pages read and rendered

83

Flutter production Dart files

107

Web TS/TSX/CSS source files

2 / 0

Mobile / web test files

Area	Inventory	Notes
Flutter source	27,418 lines; 4 data, 11 model, 12 service, 55 screen/widget/core files plus main.	Includes nine simulation templates, authentication, progress, profile, leaderboard and settings.
Mobile content/assets	20 missions, 20 text scenarios, 51 asset files.	No video dependency; three declared asset directories are absent.
Web source	9,789 lines; 35 app routes, 46 UI primitives, auth/layout, hooks, one service and two type modules.	Competency, criteria and scenario routes redirect to modules.
Server/data artifacts	0 API routes; 0 middleware; 0 schema/migration/rules files.	Mobile table names are only client references; web Firestore collections have no rules artifact.
Dependencies	Flutter: Provider, Supabase, SharedPreferences. Web: Next 15.3.8, React 19, Firebase 12, Recharts.	No web test framework, Firebase Admin SDK, Supabase web package, AI SDK or upload/storage service.
Provider configuration	Web code expects NEXT_PUBLIC_FIREBASE_*; supplied environment contract declares Supabase names.	Configuration and implementation are inconsistent.

Objectives of the Study Traceability Audit

Objective	Mobile evidence	Web evidence	Backend evidence	Overall	Major gaps
Objective 1 Scenario-based 2D simulation modules	20 missions and 20 text scenarios; five interaction types in mission_launcher.dart and simulation/templates.	Hard-coded mobile content; no instructor authoring/publish workflow or approved competency source.	No shared activity/evidence service or schemas.	PARTIAL Audit score 55%	20 permanent target/zoom/content-management gaps
Objective 2 Points, levels, rewards, achievements and progress	XP, points, levels, badges, achievements, leaderboard and progress services/screens.	Dashboard leaderboard/analytics are dummy; no approved gamification configuration.	Supabase names referenced; local and remote state compete.	PARTIAL Audit score 55%	Unapproved rules, split truth and limited tests
Objective 3 Competency modules for installation, configuration, troubleshooting, maintenance and networking	All topical areas appear in four local COCs and 20 mission titles.	Competency/criteria routes redirect; no TESDA source/version governance.	No competency, rubric or version schema in repository.	PARTIAL Audit score 30%	Topical coverage is not TESDA traceability
Objective 4 Automated skill evaluation against predefined TESDA criteria	Client templates calculate totals and store some task responses.	No attempt evidence review, score adjustment, finalization or release.	No deterministic criterion engine; client-trusted totals; no rubric version/idempotency.	INCORRECT Audit score 10%	Core evaluation lifecycle and TESDA basis absent
Objective 5 Performance analytics for progress, competency, scores, results and performance	Learner progress screen reads local/Supabase state.	Dashboard, analytics, progress detail and reports use dummy/random data.	No shared aggregates/reconciliation/report snapshots.	INCORRECT Audit score 15%	Instructor analytics do not reflect real learners

Research conclusion. None of the five objectives is fully demonstrable as an integrated, tested system. Objective 1 has the strongest implementation evidence. Objectives 4 and 5 are acceptance-blocking because evaluation is not TESDA-versioned or criterion-level and instructor analytics use fabricated data.

Mobile Application Audit

The mobile app is the most developed part of the repository. It offers a coherent learner shell and interactive mission catalog, but authority-sensitive decisions are made on the client, course/class context is absent, and assessment evidence is insufficient for the PRD evaluation lifecycle.

Area	Current implementation	Status	Gap
1. Current Architecture	Provider-based Flutter app; direct Supabase services plus SharedPreferences.	PARTIAL	No authoritative shared API; duplicated local/remote rules.
2. Authentication	AuthService uses Supabase email/password.	INCORRECT	Sign-in/splash/auth wrapper do not enforce learner role or active status.
3. Learner Profile	Profile/settings screens and Supabase profile service.	PARTIAL	No classes, instructor, membership or authoritative account-state presentation.
4. Learning Modules	Four COCs and 20 locally defined missions.	PARTIAL	Not instructor-published/assigned or versioned.
5. TESDA Competencies	COC labels and topic names only.	MISSING	No approved source, criteria or rubric mapping.
6. COC / Module Progression	Lock/completion states; first mission unlocked.	INCORRECT	50% cross-COC unlock is invented; no configured prerequisites/bypass.
7. Scenario-Based Simulation	20 scenario briefs launch 2D activities.	PARTIAL	No content version/publish state; scenario depth varies.
8. Interaction Types	Identification, drag/drop, configuration, procedure and troubleshooting.	COMPLETE	Meets the at-least-three interaction acceptance summary.
9. Drag-and-Drop	Multiple working drag/drop templates.	INCORRECT	Permanent labeled/bordered/dashed target areas remain visible.
10. Text/Video Scenarios	Text scenarios for all 20 missions.	PARTIAL	No video model, package, player or resource delivery.
11. Quiz/Knowledge Activities	Hard-coded identification/multiple-choice questions.	PARTIAL	Not instructor-managed quizzes; no answer-key/rubric version.
12. Gamification	XP, points, badges, achievements, levels and leaderboard code.	PARTIAL	Unapproved rules; split storage; runtime tables unverified.
13. Points	Separate points fields and awards.	PARTIAL	No approved point configuration or audit.
14. Levels	Database level lookup with local fallback thresholds.	PARTIAL	Fallback duplicates policy and is not versioned.
15. Badges/Achievements	Models, services and UI sections exist.	UNVERIFIED	External tables/policies/seed data not supplied.
16. Rewards	XP/points/badges function as rewards.	PARTIAL	No approved reward rules or instructor/admin governance.
17. Progress Tracking	Mission/COC progress, attempts and best/latest scores.	PARTIAL	No class assignment, rubric or final-release context.
18. Automated Evaluation	Templates compute scores immediately.	INCORRECT	Client UI logic substitutes for a criterion evaluation service.
19. Rule-Based Evaluation	Deterministic hard-coded checks in templates.	INCORRECT	No approved/versioned criteria; some rules test final selection, not actual sequence.
20. Score Calculation	Accuracy/completion/sequence/safety/time formulas.	INCORRECT	Unvalidated weights and assumed safety points.
21. Competency Status	75% maps to competent/passed.	INCORRECT	No approved TESDA source or instructor finalization.
22. Real-Time Feedback	Immediate correct/incorrect feedback and result screens.	PARTIAL	Policy for withheld feedback is unresolved; no criterion breakdown.
23. Assessment Results	Result screen and Supabase result/task inserts.	PARTIAL	No provisional/final/release state; enhanced flows duplicate writes.
24. Learner Analytics	Progress screen, recent results, achievements and XP.	PARTIAL	Mixed local/remote truth and no class/competency authorization.
25. Data Persistence	Supabase writes plus SharedPreferences resume/gamification.	INCORRECT	Split sources, multi-write completion, no idempotency/transaction.
26. Backend Integration	Direct Supabase client calls.	INCORRECT	No shared backend with web; schema/RLS unavailable.

Area	Current implementation	Status	Gap
27. Offline/Online Behavior	Local resume and local reward state.	PARTIAL	No durable offline submission queue/conflict policy.
28. Navigation	Named routes, main navigation and mission launcher.	PARTIAL	app_routes.dart is unused; role/state route checks absent.
29. Error Handling	Try/catch, messages and local-first fallback.	PARTIAL	Some failures only log; partial remote writes can remain.
30. UI/UX	Coherent theme, feedback cards, responsive scrolling.	PARTIAL	Zoom, full-screen, hidden targets and visual balance gaps.
31. Accessibility	SafeArea and touch layouts.	MISSING	No Semantics/semanticLabel/Tooltip evidence; no zoom.
32. Performance	Reasonable local data and asset-based UI.	UNVERIFIED	No profiling, device results or three-second evidence.
33. Security	Supabase Auth and anon client configuration.	INCORRECT	Role/status not enforced in app; learner supplies authoritative scores; RLS absent from repo.
34. Tests	Two test files.	MISSING	Only smoke coverage; DB test mutates a live project; no critical feature tests.

High-Value Mobile Evidence

- `lib/data/missions_data.dart`: 20 missions; all carry locally hard-coded 75 passing scores and XP rewards.
- `lib/screens/simulation/mission_launcher.dart`: dispatches five interaction templates across COC1-4.
- `lib/services/result_service.dart::saveMissionResult`: inserts client-supplied totals and task results directly to Supabase.
- `lib/screens/simulation/result_screen.dart::_saveResult`: calls completion again after enhanced templates already persisted results, enabling duplicate records.
- `lib/services/mission_service.dart::isMissionUnlocked`: unlocks the next COC at 50% previous completion without an approved requirement source.
- `lib/services/auth_service.dart`: functional Supabase session calls, but login/wrappers do not enforce learner role or active status.

Instructor Web Dashboard Audit

The visible dashboard resembles an instructor/admin console, but it is not a separate Instructor workspace and most domain screens operate on constant arrays or presentation-only state. No Instructor P0 workflow is complete end-to-end.

Capability	Current implementation	Status
Authentication	Temporary localStorage login; Firebase disabled.	INCORRECT
Authorization	Combined instructor_admin role; guard always renders.	INCORRECT
Dedicated workspace	Single combined navigation.	INCORRECT
Dashboard	Rich cards/charts over dummy arrays.	INCORRECT
Class management	No route/model/service.	MISSING
Student enrollment	No enrollment or history.	MISSING
Student monitoring	Progress list/detail use constants.	INCORRECT
Student deactivation	Local state toggle on combined user list; no reason/persistence.	INCORRECT
Student performance view	Detailed dummy profile.	INCORRECT
Module assignment	No class assignment.	MISSING
Simulation/module monitoring	Hard-coded module details; no live attempts.	PARTIAL
COC management	Hard-coded display; competency routes redirect.	PARTIAL
COC bypass	No data/UI/service/audit.	MISSING
Quiz creation/review	No quiz feature.	MISSING
AI quiz generation	No provider/draft/review flow.	MISSING
File/resource upload	No learning-resource upload.	MISSING
File management	No metadata/access/version/storage.	MISSING
Assessment viewing	Recent-result tables are dummy.	INCORRECT
Automated score viewing	Dummy totals only.	INCORRECT
Score review/adjustment	No operational flow.	MISSING
Final score/remarks/reason	No model or UI.	MISSING
Score audit history	Generic logs do not record revisions.	MISSING
Learner/competency progress	Visuals are dummy; mobile data is not shared.	INCORRECT
Analytics	DUMMY_LEARNERS/MISSIONS and random trend.	INCORRECT
Reports/export	CSV exports hard-coded rows.	INCORRECT
Search/filter/sort	Client search/filter on mock arrays; analytics COC filter unused.	PARTIAL
Permissions	No class scope or backend enforcement.	MISSING
Restricted actions	Instructor can reach user deletion/settings through combined workspace.	INCORRECT
Publishing workflow	Edit form is UI-only; create redirects; no version validation.	PARTIAL
Audit view	Firestore log viewer exists, but attribution/coverage is insufficient.	PARTIAL

Admin Web Dashboard Audit

Admin and Instructor concepts are intentionally merged in the current types and navigation. The account-management screens are security-sensitive client prototypes; they neither create/remove corresponding authentication accounts nor enforce retention, approval or scope safeguards.

Capability	Current implementation	Status
Admin authentication	Same bypassed combined login.	INCORRECT
Admin authorization/workspace	No admin role or distinct route tree.	INCORRECT
System dashboard	Dummy learner/score charts, no health/storage.	INCORRECT
User/instructor/student management	Combined list with dummy data; create uses mock UID.	INCORRECT
Account activation/deactivation	Local-only toggle/list edit; no reason or login enforcement.	INCORRECT
Permanent deletion	Deletes Firestore document client-side only; no Auth/retention safeguards.	INCORRECT
Role assignment	Only instructor_admin/learner; browser update.	INCORRECT
Permission/instructor scope	No model or service.	MISSING
Internal support	No locked-account/role-repair workflow.	MISSING
Global settings	Direct Firestore settings form; switches not demonstrably enforced.	PARTIAL
TESDA source versions	No model/configuration.	MISSING
Audit records	Generic log list and partial log writes.	INCORRECT
Security controls	UI toggles for 2FA/session timeout only.	INCORRECT
Backup/recovery	No implementation.	MISSING
System analytics/reports	Dummy shared pages.	INCORRECT
AI service configuration	No implementation; P1.	MISSING

Backend / API Audit

Finding	Evidence	Status / impact
No application backend/API layer	No <code>src/app/api</code> , server, controller, route handler or middleware files.	MISSING Secure class/content/evaluation/score/file/report operations cannot be enforced.
Mobile direct database access	<code>SupabaseConfig.client</code> used by 12 services; <code>ResultService</code> inserts client-calculated score records.	INCORRECT Trust boundary is on the learner device unless external RLS/functions compensate.
Web generic client CRUD	<code>firebase.service.ts</code> accepts arbitrary collection names and exposes <code>create/update/delete</code> in browser code.	INCORRECT No repository authorization/validation evidence.
Provider and contract split	Mobile uses Supabase tables; web uses Firebase collections and mock arrays; environment variables describe Supabase.	INCORRECT No shared record reconciliation.
UI-only/mockd functionality	Dashboard, analytics, progress, learner detail, reports, modules and users contain <code>DUMMY_*</code> or local arrays.	INCORRECT Screens cannot demonstrate operational requirements.
Validation	Web form validation is browser-side; mobile scoring and progress updates are client-side; no server schemas.	MISSING Sensitive mutations lack authoritative validation.
Idempotency/transactions	No idempotency key or unique final-attempt guard; completion spans multiple independent writes.	INCORRECT Duplicate/partial results are possible.

Database and Data Model Audit

Sixteen Supabase table/view names are referenced by the mobile client: profiles, coc_modules, missions, simulation_tasks, learner_progress, mission_results, task_results, levels, badges, user_badges, achievements, user_achievements, leaderboard_entries, view_leaderboard_overall, notifications, user_settings. No definitions, constraints, migrations, functions or RLS policies are supplied.

Domain	Evidence	Status	Capability gap
Users/profiles	Supabase profiles referenced; Firestore users referenced separately.	PARTIAL	No common user authority/schema.
Roles/account states	role/status fields in clients.	INCORRECT	Merged web role; no provider-wide enforcement/history.
Permissions/instructor scope	No records found.	MISSING	Required for RBAC and class scope.
Classes	No records found.	MISSING	Class lifecycle unsupported.
Memberships/enrollments	No records found.	MISSING	Assignments and monitoring cannot scope.
TESDA source versions	No records found.	MISSING	Traceability/reproducibility blocked.
Competencies/COCs	coc_modules referenced; web COC arrays.	UNVERIFIED	No definition/source/version.
Modules/versions	missions referenced and hard-coded.	PARTIAL	No immutable published versions.
Scenarios	Hard-coded Dart scenario data.	PARTIAL	No content record/version/media.
Activities/tasks	simulation_tasks and task_results referenced.	PARTIAL	No schema/criterion linkage.
Attempts	mission_results used as attempt-like rows.	PARTIAL	No explicit submission state/idempotency/rubric version.
Learner action events	No chronological event entity found.	MISSING	Sequence/evidence cannot be reconstructed.
Evaluation criteria/rubrics	No records found.	MISSING	Objective 4 blocked.
Automated evaluations	Totals embedded in result rows/client.	INCORRECT	No criterion-level engine/output.
Scores	score/percentage/pass fields exist.	PARTIAL	Client supplied; no authority/version.
Final scores/release	No records found.	MISSING	No provisional/final distinction.
Score overrides/reasons	No records found.	MISSING	Panel #7 blocked.
Audit logs	Separate Firestore logs referenced.	INCORRECT	Not shared, complete or reliably attributed.
Points/levels/achievements	Eight gamification tables/views referenced.	UNVERIFIED	Definitions and policy unavailable.
Progress	learner_progress referenced.	PARTIAL	No class/assignment/final-result context.
Quizzes/questions/attempts	No database model found.	MISSING	Instructor quiz workflow absent.
Files/resources	No database/storage model found.	MISSING	Upload/access governance absent.
Reports/analytics snapshots	No model found.	MISSING	Web reports use constants.
Constraints/RLS/migrations	No schema, migrations, Firestore rules or Supabase policies in repository.	UNVERIFIED	Integrity/security cannot be proven.

Verification boundary. Client references prove expected names, not that tables, relationships, constraints, policies or triggers exist. Database capabilities that depend on external Supabase/Firebase configuration remain UNVERIFIED.

Authentication & RBAC Audit

Permission Matrix

Feature	Learner	Instructor	Admin	Current implementation	Expected	Status
Log in	Assigned active account	Assigned active account	Assigned active account	Web bypass; mobile Supabase session only	Role/state checked by backend	INCORRECT
Account creation	No	No	Yes	Mobile self-signup; public web privileged signup; combined user form	Admin governed; approved pending instructor flow if retained	INCORRECT
Class creation	No	Owned scope	Govern/read	Absent	Instructor class lifecycle	MISSING
Enrollment	View own	Manage owned class	Govern/read	Absent	Class membership/history	MISSING
Student monitoring	Own	Assigned classes	System-wide read	Mobile own progress; web dummy global data	Backend-scoped real data	INCORRECT
Modules/content	Consume published	Author/publish	Govern/disable	Mobile hard-coded; web hard-coded/edit UI-only	Versioned publish workflow	PARTIAL
Assessment attempts	Perform/view own	Review assigned	System-wide read/audit	Mobile direct writes; web dummy	Immutable evidence and scoped read	INCORRECT
Score override/finalize	No	Yes, audited	Emergency only	Absent	Revision/reason/final/r release	MISSING
COC bypass	No	Yes, audited	Configure/audit	Absent	Access-only audited bypass	MISSING
Student deactivation	No	Assigned learner + reason	Yes	Combined local toggle; generic edit	Backend state, reason and preserved history	INCORRECT
Permanent account deletion	No	No	Admin only	Combined page client-deletes Firestore doc	Admin-only guarded retention workflow	INCORRECT
Instructor deletion	No	No	Admin only	Same combined page	Admin-only safeguards	INCORRECT
File management	Consume authorized	Manage class resources	Govern	Absent	Validated scoped storage	MISSING
AI quiz drafts	No	Draft/review/publish	Configure service	Absent	P1 instructor-reviewed only	MISSING
Reports	Personal	Assigned scope	System-wide	Mobile personal partial; web dummy	Source-reconciled scoped exports	INCORRECT
System configuration	No	No	Yes	Visible to combined role; browser update	Admin-only enforced settings	INCORRECT
Audit logs	Own if shown	Own instructional	System-wide	Generic global log list in combined workspace	Role-scoped immutable audit	INCORRECT

Critical Role Findings

- `src/types/index.ts` explicitly comments that Instructor/Admin are merged; only `instructor_admin` | `learner` exist.
- `LoginForm` accepts any email containing `@` and any non-empty password, then sets `bytequest_temp_auth`.
- `ProtectedRoute` always renders children; Firebase session enforcement is commented out.
- Public web sign-up creates an active `instructor_admin` Firebase/Firestore user with no Admin approval.
- Mobile login validates only Supabase credentials; `isActiveAccount()` exists but is not called by sign-in, splash or auth wrapper.

TESDA Alignment Audit

Required control	Current evidence	Status
Approved booklet/training-regulation version	No TESDA source/version model or configuration. PRD OD-01 remains open.	MISSING
COC/competency structure	Four locally named COCs and 20 missions in mobile code; web has hard-coded COC arrays and redirect-only competency routes.	PARTIAL
Criterion and rubric mapping	Type stubs exist in web; no criterion records, CRUD, evaluator or attempt linkage.	MISSING
Competency score vs points	Separate fields exist, but both are calculated/awarded by unapproved client rules.	PARTIAL
Passing/grading logic	Mobile model uses 75%; web uses 75/80/85 and presentation bands. No source validation.	INCORRECT
Traceable activity/score mapping	Mission and COC IDs are stored, but no rubric version or affected criteria.	PARTIAL
Instructor decision record	No score review, finalization, adjustment or release model.	MISSING

Mandatory wording: TESDA value/rule requires validation against the approved TESDA CSS NC II booklet.

SECTION 17

Automated Skill Evaluation Audit

The app provides automated totals and immediate feedback, but this is not the PRD's versioned, criterion-level, auditable evaluation service.

Evaluation concern	Current behavior	Gap
Expected procedure	Hard-coded task lists/questions and local validation in simulation templates.	No approved rubric/version; rules are embedded in UI code.
Action tracking	Some task rows store answer/target, correctness, attempts and hints.	No comprehensive chronological event stream, criterion match or submission state.
Sequence checking	<code>step_procedure_mission_screen.dart</code> validates that earlier steps are selected, not the order in which actions occurred.	Chronology can be misclassified as correct.
Score calculation	Several templates use 40% accuracy + 20% completion + 20% sequence/correctness + 10% assumed safety + time bonus.	Embedded, unversioned and unvalidated weights; safety can be awarded without evidence.
Competency rating	<code>MissionResultDb</code> maps 75% to competent/passed.	TESDA/instructor source not supplied or linked.
Stored result	Mission and task result inserts are attempted in Supabase.	Client trust, no idempotency, duplicate enhanced-screen path and no immutable final record.
Instructor review	No operational web flow.	No adjustment, reason, affected criteria, finalization, release or revision history.

SECTION 18

Gamification Audit

Element	Evidence	Status
Points	<code>pointsReward</code> , <code>profile total_points</code> , <code>result earned_points</code> .	PARTIAL separate fields, unapproved rules.
Levels	<code>GamificationService.getLevelFromDb</code> plus hard-coded fallback thresholds.	PARTIAL duplicated formulas.
Badges/achievements	Models and Supabase service queries for badges, achievements and user awards.	UNVERIFIED tables/rules/runtime unavailable.
Rewards	XP/points/badges and achievement XP exist; no separate approved reward catalog/workflow.	PARTIAL
Progress	Mission/COC progress screens, best/latest score and attempts fields.	PARTIAL class/competency truth and history incomplete.
Leaderboard	Ranking formula $XP \times 10 + \text{missions} \times 100$; web leaderboard redirects to analytics.	PARTIAL unapproved and disconnected.
Persistence integrity	Local and Supabase gamification services both update state; local reconciliation keeps maximum values.	INCORRECT competing sources of truth.

SECTION 19

Analytics & Reporting Audit

Learner-facing: the mobile progress screen reads a mixture of Supabase and local state and shows mission/COC completion, results, achievements and XP. This is useful but not tied to class assignments or approved competency criteria.

Instructor-facing: `src/app/dashboard/page.tsx`, `analytics/page.tsx`, `progress/*` and `reports/page.tsx` use dummy learners, mission results, random trends and fixed error areas. Report parameters do not query source records; CSV exports the

constants.

Objective 5 impact. The current web analytics cannot serve as evidence of actual learner monitoring. Visual completeness must not be confused with data correctness.

SECTION 20

Class & Enrollment Management Audit

MISSING No class route, model, service, collection/table reference, enrollment, membership history, assignment or instructor ownership implementation was found. Mobile profile/course screens do not display enrolled classes, assigned instructor, availability windows or class-specific modules.

This blocks MOB-002, INS-002/003/009/011/017, ADM-004/008, SHR-002 and Panel Recommendation 9. It also prevents reliable authorization scoping and real class analytics.

SECTION 21

Quiz & AI Assistance Audit

MISSING No web quiz route, data model, item editor, answer key, rubric mapping or publish/review state exists. Mobile identification questions provide knowledge checks, but they are hard-coded activities rather than instructor-managed quizzes.

No generative AI provider, SDK, server credential boundary, draft provenance or instructor approval workflow exists. This is a P1 gap and must remain deferred until P0 role, data, rubric and publishing controls are stable.

SECTION 22

File / Learning Resource Management Audit

MISSING No learning-resource file model, upload service, storage path, validation rule, access scope, metadata, version or deletion workflow exists. The only web file input is the user-import page, which does not parse the chosen CSV and substitutes three hard-coded preview rows.

Consequently, PDF/image/video supporting materials, limits/content types, authorized streaming/download and file audit records cannot be verified.

UI/UX Audit

Area	Positive evidence	Issue / status
Navigation/role clarity	Consistent side/top navigation and mobile bottom navigation.	INCORRECT Web labels “Instructor/Admin” and “Insights & Admin”; roles are not separated.
Responsive behavior	Tailwind breakpoints, mobile sheet sidebar, Flutter SafeArea and scrolling.	PARTIAL runtime device matrix unverified.
Loading/error/empty states	Skeletons, spinners, toasts and several no-data messages.	PARTIAL mock delays mask real service behavior; some catches only log.
Simulation visibility	Large illustrative work areas.	MISSING no InteractiveViewer, zoom, full-screen or landscape; app locks portrait.
Drag-and-drop clarity	Feedback and labeled components.	INCORRECT permanent labeled/bordered target zones and six dashed rectangles remain visible.
Visual hierarchy	Generally coherent palette/cards.	PARTIAL large 200px/140px mascot/logo treatments compete with content.
Accessibility	Many controls have labels; responsive touch layouts.	PARTIAL no Flutter Semantics/semanticLabel/Tooltip evidence; only sparse web ARIA; no zoom.
Encoding/polish	Professional visual theme.	PARTIAL mojibake appears in UI/source strings (for example the command-key label and bullets).

Security Audit

Risk	Evidence	Rating
Authentication bypass	LoginForm, use-auth.tsx and ProtectedRoute explicitly disable Firebase checks and trust localStorage.	CRITICAL P0
Privilege self-registration	Public SignUpForm creates active instructor_admin accounts.	CRITICAL P0
Missing backend authorization	Generic browser Firestore CRUD and no repository rules/API; direct learner Supabase score writes; external RLS unavailable.	CRITICAL P0
Role confusion	Instructor/Admin merged in types, mock user, navigation and data screens.	CRITICAL P0
Object-level access	Client accepts arbitrary document IDs and user IDs; no class-scope server check.	HIGH P0
Score trust	Learner client sends score, percentage, passed and task values; no server recomputation evidence.	CRITICAL P0
Account deletion	Combined workspace directly deletes a Firestore user document; not the Auth account; no retention/audit safeguards.	HIGH P0
Audit attribution	logAction records display text only; Firebase auth.currentUser is null under bypass, producing "System".	HIGH P0
Credential hygiene	Mobile integration test embeds a project URL and anon JWT. An anon key is client-public by design but makes correct RLS essential and should not be duplicated in tests.	MEDIUM P1
Secret boundary	.env.local is ignored by .gitignore; no service-role value was printed or referenced by client source.	CONTROL NOTE

No vulnerability was exploited. Findings are based solely on non-destructive static inspection.

SECTION 25

Testing and QA Audit

Test area	Current coverage	Status
Web unit/integration/E2E	No test files, framework, scripts or dev dependencies.	MISSING
Mobile smoke widget	test/widget_test.dart checks app title and constructs the splash app.	PARTIAL
Mobile database test	test/db_test.dart initializes a real Supabase project, creates a user and inserts profile/results.	INCORRECT mutable integration script, no isolation/cleanup; not run in this audit.
Authentication/account state	No meaningful role/status/deactivation tests.	MISSING
Simulation/evaluation/scoring	No deterministic rule, criterion, sequence or content-answer tests.	MISSING
Progress/gamification/idempotency	Database test covers selected XP calculations, using a fallback formula that differs from production; no duplicate-submission test.	INCORRECT
Classes/scores/files/RBAC	Features absent; tests absent.	MISSING
Accessibility/performance/security	No automated or recorded manual results.	MISSING

SECTION 26

Scope & Limitation Compliance

The implementation generally stays within the approved technical boundary: 2D predefined activities, supplementary learning, Flutter/Next.js and no 3D/AR/VR, official TESDA integration, proctoring, real-device control, multi-tenancy, finance, HR or registrar functions.

Boundary	Finding	Status
2D predefined simulations	Implemented as images/cards/drag areas and deterministic prompts.	COMPLETE
Supplementary, not official certification	No certificate generation, but the required explicit disclaimer is absent.	PARTIAL
LMS-lite only	No enterprise-LMS expansion; however, the required limited classes/files/quizzes are also missing.	PARTIAL
No autonomous AI assessment	No AI exists.	COMPLETE
No official TESDA integration claim	No API integration exists; UI uses CSS NC II labels without source version.	PARTIAL
Technology continuity	Flutter and Next.js retained.	COMPLETE

Out-of-scope code found: no material ERP/SIS/finance/HR/video-conferencing expansion. Leaderboards and user CSV import remain adjacent to approved gamification/account governance; they should not outrank P0 requirements.

PRD Compliance Matrix

Every explicit MOB, INS, ADM, SHR, EVAL, UI and NFR identifier from the PRD is included. Statuses are conservative static-review judgments.

MOB Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
MOB-001	Authenticate learner; enforce active authorized account	Mobile	auth_service.dart; auth_wrapper.dart; splash_screen.dart	INCORRECT	Role/status checks not used; self-signup	P0
MOB-002	Profile, classes, instructor, assigned modules, availability	Mobile	profile_screen.dart; no class/enrollment code	PARTIAL	Profile only	P0
MOB-003	Configured COC sequence and lock states	Mobile	mission_service.dart::isMissionUnlocked	INCORRECT	Invented 50% rule; no bypass state	P0
MOB-004	Text and/or video scenarios	Mobile	mission_scenarios_data.dart (20 text)	PARTIAL	No video support	P0
MOB-005	Three or more non-static interaction types	Mobile	mission_launcher.dart; five templates	COMPLETE	No material gap for variety	P0
MOB-006	No permanent drag target boxes in assessment	Mobile	drag_drop_mission_screen.dart; coc1_m2_screen_enhanced.dart	INCORRECT	Labeled/dashed zones always visible	P0
MOB-007	Zoom and full-screen	Mobile	No InteractiveViewer/fullscreen; portrait lock in main.dart	MISSING	No visibility controls	P0
MOB-008	Capture start/end/actions/order/errors/time/criteria/state	Mobile	result_service.dart; task_result_model.dart; resume service	PARTIAL	Order, criteria and submission state incomplete	P0
MOB-009	Criterion feedback and provisional result	Mobile/Backend	Simulation feedback; client total	INCORRECT	No criterion engine/provisional state	P0
MOB-010	Release-gated final score with clear labels	Mobile/Web	result_screen.dart	MISSING	No final/release model	P0
MOB-011	Competency/module/attempt/achievement/point progress	Mobile	progress_screen.dart; gamification services	PARTIAL	No approved competency/class context	P0
MOB-012	Instructor retry limits/windows; preserve history	Mobile/Web	No model/service	MISSING	No limits/windows/history policy	P0
MOB-013	Safe failures and duplicate prevention	Mobile/Backend	try/catch; local resume; ResultScreen duplicate path	INCORRECT	No idempotency; partial writes possible	P0
MOB-014	Balanced visuals and minimized logos	Mobile	splash/login/signup imagery	PARTIAL	Oversized imagery remains	P2
MOB-015	No unreviewed AI content for learners	Shared	No AI code; no publication workflow	PARTIAL	AI absent but review/publish guarantee absent	P1

INS Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
INS-001	Separate Instructor workspace/guard	Web	types/index.tsx; sidebar.tsx; ProtectedRoute	INCORRECT	Merged role and no guard	P0
INS-002	Class create/edit/archive/view	Web/Backend	No class route/model	MISSING	Entire lifecycle absent	P0
INS-003	Enrollment and membership history	Web/Backend	No enrollment code	MISSING	Absent	P0
INS-004	Deactivate assigned learner with reason; no delete	Web/Backend	users/page.tsx local toggle; edit delete button	INCORRECT	No reason/scope/persistence; delete exposed	P0
INS-005	Content CRUD/preview/publish/archive mapped to COC	Web	modules hard-coded; edit TODO; redirect routes	PARTIAL	No persistence/version/publish validation	P0
INS-006	Manage PDFs/images/videos/resources	Web/Backend	No learning-resource feature	MISSING	Absent	P0
INS-007	Quiz/key/criteria/weights from approved rubric	Web/Backend	AssessmentCriteria type only; routes redirect	MISSING	No quiz/rubric implementation	P0
INS-008	Optional AI quiz drafts; instructor publishes	Web/Backend	No AI dependency/route	MISSING	P1 feature absent	P1
INS-009	Class order/prerequisites/availability	Web/Backend	Mission edit lock field UI-only	PARTIAL	No class policy persistence	P0
INS-010	Audited COC bypass; no auto-pass	Web/Backend	No bypass code	MISSING	Absent	P0
INS-011	Near-real-time class/learner monitoring	Web	progress/dashboard dummy arrays	INCORRECT	Not real or scoped	P0
INS-012	Review actions/sequence/responses/criteria	Web/Backend	Dummy learner history only	MISSING	No evidence viewer	P0
INS-013	Confirm/edit/final score, remarks, reason	Web/Backend	No score review route/model	MISSING	Absent	P0
INS-014	Release/withhold finalized results	Web/Mobile	No release field/flow	MISSING	Absent	P0
INS-015	Class/learner/module/competency reports/export	Web	reports/page.tsx dummy CSV	INCORRECT	Export not sourced/reconciled	P0
INS-016	Intervention indicators	Web	analytics hard-coded commonMistakes/support list	INCORRECT	Fabricated, not linked to evidence	P0
INS-017	Assigned-scope authorization	Web/Backend	No class/scope/server enforcement	MISSING	Cross-class access not preventable	P0

ADM Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
ADM-001	Separate Admin workspace/guard	Web	Merged instructor_admin role	INCORRECT	No distinct role/routes	P0
ADM-002	Manage accounts/roles/initial state	Web/Backend	SignUpForm; users/create with mock UID	INCORRECT	Privileged self-signup; no Auth account in admin create	P0

ID	Requirement	System	Evidence	Status	Gap	Priority
ADM-003	Activate/deactivate/reset/delete with safeguards	Web/Backend	users edit/delete client Firestore document	INCORRECT	No retention/Auth delete/audit safeguard	P0
ADM-004	Manage instructor scope/permissions	Web/Backend	No scope/permission model	MISSING	Absent	P0
ADM-005	Resolve internal account/access problems	Web/Backend	Generic user/settings UI	MISSING	No governed support workflow	P0
ADM-006	Global settings/file limits/report config/flags	Web/Backend	settings/page.tsx direct update	PARTIAL	Settings not enforced; limits/config missing	P0
ADM-007	Register/activate approved TESDA versions	Web/Backend	No TESDA source model	MISSING	Absent	P0
ADM-008	System-wide read/health/storage without teaching rights	Web	dashboard dummy; merged edit rights	INCORRECT	No health/storage; boundaries wrong	P0
ADM-009	Review comprehensive audit events	Web	logs/page.tsx; logAction helper	INCORRECT	Incomplete events and actor attribution	P0
ADM-010	Backups/recovery/export	Backend	No implementation	MISSING	Absent	P0
ADM-011	Global availability/remove corrupt content with audit	Web/Backend	No content governance service	MISSING	Absent	P0
ADM-012	System analytics/admin reports	Web	analytics/reports dummy	INCORRECT	Not source-backed	P0
ADM-013	No routine score edit; emergency audited privilege	Web/Backend	No score correction model	MISSING	Boundary absent	P0
ADM-014	Optional AI service configuration	Web/Backend	No AI code	MISSING	P1 absent	P1

SHR Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
SHR-001	One role-aware auth/authorization layer	Shared	Mobile Supabase; web bypassed Firebase	INCORRECT	Different providers/no common RBAC	P0
SHR-002	Secure services for core domains	Backend	No API/server routes	MISSING	Absent	P0
SHR-003	Stable identifiers/timestamps across products	Shared	Supabase UUID resolver; web mock IDs/Firebase docs	INCORRECT	No common identifier contract	P0
SHR-004	Preserve attempt/score/audit history	Shared	mission_results/task_results; generic logs	PARTIAL	No immutable revisions/final history	P0
SHR-005	Validate files/limits/type/scope	Backend	No resource upload service	MISSING	Absent	P0
SHR-006	Prevent duplicate submission/double scoring	Backend	No key; duplicate enhanced result path	INCORRECT	Duplicates demonstrably possible	P0
SHR-007	Version modules/criteria/TESDA source	Shared	No version model	MISSING	Absent	P0

ID	Requirement	System	Evidence	Status	Gap	Priority
SHR-008	Data export and backup	Shared	Dummy report CSV only	MISSING	No essential-data backup/restore	P0
SHR-009	Audit high-impact mutations	Shared	Partial web logAction calls	INCORRECT	Critical actions omitted/unattributed	P0
SHR-010	Compatibility with existing codebase	Shared	Flutter/Next retained	PARTIAL	Provider/data split prevents integration	P0

EVAL Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
EVAL-001	Activity links COC, rubric version and criteria	Evaluation	Mission COC IDs only	MISSING	No rubric/criteria/version	P0
EVAL-002	Deterministic, testable predefined rules	Evaluation	Hard-coded template formulas	INCORRECT	Not approved/versioned/tested	P0
EVAL-003	Record correct/incorrect/incomplete/out-of-sequence	Mobile	Some task result fields and mistakes	PARTIAL	No full chronological event evidence	P0
EVAL-004	Criterion-level results	Evaluation	Total score plus task rows	MISSING	No criterion evaluation outputs	P0
EVAL-005	Adjustment preserves all required fields	Web/Backend	No adjustment model	MISSING	Absent	P0
EVAL-006	Final immutable; correction creates revision	Backend	No final/revision model	MISSING	Absent	P0
EVAL-007	Bypass unlock only	Web/Backend	No bypass implementation	MISSING	Absent	P0
EVAL-008	Attempt retains rubric version	Backend	No version reference	MISSING	Absent	P0
EVAL-009	Learner cannot edit protected evidence/score	Security	No score-edit UI; RLS/server rules unavailable	UNVERIFIED	Backend prohibition cannot be proven	P0
EVAL-010	Supplementary/not official TESDA disclosure	All UI	No disclaimer found	MISSING	Required disclosure absent	P0

UI Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
UI-001	Zoom/full-screen	Mobile	No relevant widget/control	MISSING	Absent	P0
UI-002	No permanent assessment targets	Mobile	Visible target containers/dashed rectangles	INCORRECT	Panel correction not applied	P0
UI-003	More than drag-and-drop	Mobile	Five interaction types	COMPLETE	Satisfied	P0
UI-004	Text and video context	Mobile	Text only	PARTIAL	Video absent	P0
UI-005	Minimize oversized decorative assets	Mobile/Web	Generally compact web; large mobile art	PARTIAL	Visual priority inconsistent	P2
UI-006	Clear practice/assessment/result/lock labels	All UI	Some lock/completion labels	MISSING	Practice/assessment/provisional/final/bypass incomplete	P0

ID	Requirement	System	Evidence	Status	Gap	Priority
UI-007	Contrast/scalable text/keyboard/touch accessibility	All UI	Responsive controls; sparse ARIA; no Flutter semantics	PARTIAL	Accessibility evidence incomplete	P0
UI-008	Prevent accidental submit/finalize; recover drafts	All UI	Some dialogs/local resume	INCORRECT	Mission final submission lacks confirmation/idempotency	P0

NFR Requirements

ID	Requirement	System	Evidence	Status	Gap	Priority
NFR-001	Backend RBAC, sessions, credentials, least privilege	Security	Web bypass; no backend rules	INCORRECT	Critical security controls absent	P0
NFR-002	Submission/scoring/finalization integrity	Shared	Multi-write client workflow	INCORRECT	No atomic/transactional finalization	P0
NFR-003	Audit actor/time/old/new/reason/target	Shared	Generic log text only	INCORRECT	Required fields/events absent	P0
NFR-004	Usable in ~3 seconds; prompt scoring	All	No measurements	UNVERIFIED	Deployment performance untested	P1
NFR-005	Graceful errors/retry-safe/no silent loss	All	Try/catch/local resume	INCORRECT	Duplicate and partial remote state possible	P0
NFR-006	Consistent usable navigation/status labels	All UI	Coherent navigation/theme	PARTIAL	Role/result-state ambiguity	P1
NFR-007	Accessibility controls	All UI	Responsive/touch basics	PARTIAL	No zoom/semantics/recorded audit	P1
NFR-008	Browser and existing mobile target compatibility	Platforms	Scaffolds exist	UNVERIFIED	No current build/device results	P1
NFR-009	Typed/modular/validated/tested maintainability	Codebase	Services/types exist; many any/dummy/duplicates	PARTIAL	Tests/config docs insufficient	P1
NFR-010	Privacy-scoped learner records/reports	Security	No class scope; dummy personal records	INCORRECT	Authorization/privacy not enforced	P0
NFR-011	Periodic backup/tested restore/export	Backend	No implementation	MISSING	Absent	P0
NFR-012	Operational logging/diagnostics	All	console/debugPrint and partial logs	PARTIAL	No structured service observability	P1
NFR-013	Historical content/rubric immutability	Shared	No versioning	MISSING	Absent	P0
NFR-014	Repeatable rule/permission tests	QA	No critical tests	MISSING	Absent	P0

Panel Recommendation Compliance Matrix

#	Recommendation	Current evidence	Status	Missing / issue	Priority
1	TESDA-based grading/modules	Local COCs and hard-coded passing thresholds only	MISSING	No approved source/version/criteria; conflicting 75/80/85 values	P0
2	Separate Instructor; Admin internal	types/index.ts merges roles; single sidebar/workspace	INCORRECT	No separate roles, guards or APIs	P0
3	Instructor bypass COC 1-4	No bypass code	MISSING	No scope/reason/date/identity/audit; no access-only rule	P0
4	Simulation visibility/zoom	Portrait lock; no InteractiveViewer/fullscreen	MISSING	No demonstration controls	P0
5	Remove drag/drop boxes	Installation Areas and dashed drop markers visible	INCORRECT	Assessment reveals target zones	P0
6	TESDA-aligned points/scoring	Client hard-coded scores, XP and ranking formula	INCORRECT	No approved configured structure	P0
7	Instructor review/edit/finalize task score	Only dummy score tables	MISSING	No evidence, override, reason, final or history	P0
8	Optional instructor AI quiz drafts	No quiz or AI feature	MISSING	P1 draft/review/provenance boundary absent	P1
9	LMS-style classes/enrollment	No class/enrollment implementation	MISSING	Entire workflow absent	P0
10	Instructor files and monitoring	No resource upload; monitoring is dummy	MISSING	Storage/access/live learner data absent	P0
11	Instructor deactivates; Admin deletes	Combined local toggle and client delete	INCORRECT	No reason/scope/retention/Admin-only enforcement	P0
12	Text or video scenarios	20 text scenarios	PARTIAL	Video absent	P0
13	More interactive than drag/drop	Five interaction types across all COCs	COMPLETE	Variety met; quality/criteria still need validation	P0
14	Minimize logos/figures	Compact web chrome; large mobile splash/login art	PARTIAL	Mobile visual hierarchy needs validation	P2

Objective-to-Implementation Matrix

Objective	Implemented evidence	What prevents full achievement	Current judgment	Audit score
Objective 1 - Scenario-based 2D simulation modules	Mobile: 20 missions and 20 text scenarios; five interaction types in mission_launcher.dart and simulation/templates. Web: Hard-coded mobile content; no instructor authoring/publish workflow or approved competency source. Backend: No shared activity/evidence service or schemas.	20 permanent target/zoom/content-management gaps	PARTIAL	55%
Objective 2 - Points, levels, rewards, achievements and progress	Mobile: XP, points, levels, badges, achievements, leaderboard and progress services/screens. Web: Dashboard leaderboard/analytics are dummy; no approved gamification configuration. Backend: Supabase names referenced; local and remote state compete.	Unapproved rules, split truth and limited tests	PARTIAL	55%
Objective 3 - Competency modules for installation, configuration, troubleshooting, maintenance and networking	Mobile: All topical areas appear in four local COCs and 20 mission titles. Web: Competency/criteria routes redirect; no TESDA source/version governance. Backend: No competency, rubric or version schema in repository.	Topical coverage is not TESDA traceability	PARTIAL	30%
Objective 4 - Automated skill evaluation against predefined TESDA criteria	Mobile: Client templates calculate totals and store some task responses. Web: No attempt evidence review, score adjustment, finalization or release. Backend: No deterministic criterion engine; client-trusted totals; no rubric version/idempotency.	Core evaluation lifecycle and TESDA basis absent	INCORRECT	10%
Objective 5 - Performance analytics for progress, competency, scores, results and performance	Mobile: Learner progress screen reads local/Supabase state. Web: Dashboard, analytics, progress detail and reports use dummy/random data. Backend: No shared aggregates/reconciliation/report snapshots.	Instructor analytics do not reflect real learners	INCORRECT	15%

Objective scores use a five-part evidence rubric: user-visible capability, persistence, backend authority, traceability and tests. Each part contributes 20 points. Partial evidence receives proportional credit; dummy data receives no persistence/backend credit.

Complete Features

Interaction diversity

The mobile launcher implements five distinct interactive patterns and distributes them across COC1-4, exceeding the PRD minimum of three.

Evidence: ByteQuest Mobile App/lib/screens/simulation/mission_launcher.dart

Text scenario coverage

All 20 locally defined missions have scenario context, titles, objectives and narrative prompts.

Evidence: ByteQuest Mobile App/lib/data/mission_scenarios_data.dart

Core 2D scope

Activities remain two-dimensional and predefined; no unsupported 3D/AR/VR or real-device control was introduced.

Evidence: simulation/templates and mobile asset inventory

Technology continuity

The existing Flutter and Next.js/TypeScript stacks are retained.

Evidence: ByteQuest Mobile App/pubspec.yaml; package.json

No autonomous AI assessment

No AI dependency or learner grading/publishing path exists.

Evidence: package.json, pubspec.yaml and source-wide AI search

Partially Implemented Features

Mobile learner shell

Authentication, profile, dashboard, course/mission map, results, progress, leaderboard and settings screens exist, but class context and account enforcement are incomplete.

Evidence: ByteQuest Mobile App/lib/screens

Attempt/result persistence

Mission and task records are inserted into Supabase, but evidence/version/finality/idempotency are incomplete and policies are unavailable.

Evidence: result_service.dart; mission_completion_service.dart

Gamification

Points, XP, levels, badges, achievements and rankings have models/services/UI, but rules and truth ownership are not approved.

Evidence: gamification_service.dart; local_gamification_service.dart; leaderboard_service.dart

Web content and monitoring shells

Modules, learner monitoring, analytics, reports and settings have usable presentation layouts, but most are hard-coded or UI-only.

Evidence: src/app/modules, progress, analytics, reports, settings

Error and resume behavior

Mobile templates save local resume state and use catch paths; remote writes are not atomic and duplicate-safe.

Evidence: progress_resume_service.dart; result_screen.dart

SECTION 32

Missing Features

Shared backend and common identity

No role-aware API/service layer connects mobile and web.

Evidence: No src/app/api, middleware, server code or shared provider

Classes, enrollment and assignments

No lifecycle, membership, history, instructor ownership or mobile class view.

Evidence: No class/enrollment/assignment route/model/service

TESDA source, rubrics and criteria

No approved edition, criteria, weights or immutable version linkage.

Evidence: No TESDA/rubric/criterion records; redirect-only web routes

Instructor assessment lifecycle

No evidence review, adjustment reason, final score, revision or release.

Evidence: No score review/finalization routes/models

Learning resources and quizzes

No governed file or instructor quiz workflow; no optional AI drafts.

Evidence: No learning-resource/quiz/AI code

Backup, restore and versioned history

No recovery process, report snapshots or immutable content/rubric revisions.

Evidence: No backup/version schema or service

SECTION 33

Incorrect Implementations

Web authentication and route protection

Authentication is explicitly bypassed and protected content renders without a user.

Evidence: LoginForm.tsx; use-auth.tsx; protected-route.tsx

Role model

Instructor and Admin are merged, conflicting with the primary role boundary.

Evidence: src/types/index.ts; src/types/auth.ts

Scoring authority

The learner client awards unvalidated scores, competence and rewards.

Evidence: simulation templates; MissionResultDb; ResultService

Analytics/report truth

Web monitoring and reports are calculated from dummy/random values.

Evidence: dashboard/page.tsx; analytics/page.tsx; progress/*; reports/page.tsx

Account governance

Combined UI locally toggles users and directly deletes a Firestore document without Auth/retention safeguards.

Evidence: users/page.tsx; users/[id]/edit/page.tsx

Drag target and result integrity

Assessment targets remain visible and enhanced flows save completion twice.

Evidence: drag_drop_mission_screen.dart; coc1_m2_screen_enhanced.dart; result_screen.dart

SECTION 34

Unverified Implementations

Supabase schema and RLS

Sixteen names are referenced, but table definitions, constraints, triggers and policies are external.

Evidence: Mobile service .from(...) calls; no schema/migrations

Firestore project and rules

Client initialization exists, but supplied environment names do not match and no rules artifact is present.

Evidence: src/lib/firebase.ts; .env.example

Badges/achievements/leaderboard runtime

Models and service calls exist; live data and policy behavior were not executed.

Evidence: gamification_service.dart; leaderboard_service.dart

Build and platform compatibility

Static scaffolds exist, but no build/type-check/device/browser execution was performed under the read-only mandate.

Evidence: pubspec.yaml; platform manifests; package.json

Performance and resilience

No deployment metrics, network-failure trial or concurrency test results were supplied.

Evidence: No performance/test artifacts

SECTION 35

Bugs / Technical Risks

Risk	Evidence	Priority
Duplicate mission persistence	Enhanced identification/COC1 M2/M3 templates call MissionCompletionService before navigation; ResultScreen.initState calls it again.	P0
Impossible RJ45 answer	Question option is 'RJ45 Connector' while correctAnswer is 'RJ45', so exact comparison cannot succeed.	P0
False sequence validation	Procedure validator examines the final selected set, not chronological action order.	P0
Assumed safety points	Several score formulas award 10 safety points without a recorded safety action.	P0
Conflicting pass thresholds	Mobile catalog/model uses 75; web module arrays display 75/80/85; neither is approved.	P0
Provider/config mismatch	Web code needs Firebase environment values while .env.example/.env.local define Supabase values.	P0
Mock UID account creation	Web admin create writes a Firestore record with Math.random UID and ignores the password field for Auth.	P0
CSV import is not parsing	Any selected CSV yields three hard-coded learners; stated duplicate/size controls are not implemented.	P1
Unused analytics filters	Selected COC/report selectors do not filter the constant source arrays.	P1
Dual Android Gradle files	build.gradle and build.gradle.kts coexist with different Java/SDK settings; release uses debug signing.	P1
Declared assets absent	pubspec declares assets/icons, animations and illustrations directories that are not present.	P1
Mojibake strings	Several UI/debug strings contain malformed encoded characters.	P2

SECTION 36

Security / Permission Risks

ID	Risk	Severity	Priority
SEC-01	Web session is a forgeable localStorage flag	Critical	P0
SEC-02	Public sign-up grants active instructor_admin	Critical	P0
SEC-03	No distinct Instructor/Admin authorization	Critical	P0
SEC-04	Generic client-side collection CRUD	Critical	P0
SEC-05	Client-supplied learner score/pass/competence	Critical	P0
SEC-06	No class/object-level scope enforcement	High	P0
SEC-07	Mobile active/learner state not checked	High	P0
SEC-08	Deletion bypasses retention and Auth account lifecycle	High	P0
SEC-09	Audit actor and high-impact coverage insufficient	High	P0
SEC-10	File upload validation/access controls absent	High	P0
SEC-11	RLS/Firestore rules unavailable	Unverified high	P0
SEC-12	Project anon key duplicated in mutable test	Medium	P1

SECTION 37

UI/UX Issues

ID	Issue	Priority
UX-01	No zoom/full-screen; portrait-only simulation	P0
UX-02	Permanent drag target boxes/dashed markers	P0
UX-03	No video scenario player	P0
UX-04	No practice vs assessment or provisional vs final labels	P0
UX-05	Large mobile mascot/logo elements	P2
UX-06	Sparse semantic/keyboard accessibility evidence	P1
UX-07	Mock data creates false success/empty/loading states	P1
UX-08	No submission confirmation for authoritative attempt	P1
UX-09	Encoding artifacts in visible strings	P2

SECTION 38

Testing Gaps

Area	Current test evidence	Priority
Authentication and account state	None meaningful	P0
RBAC and route/API authorization	None	P0
Mobile navigation/session role	Smoke only	P0
Simulation interactions/content answers	None	P0
Deterministic evaluation/criteria	None	P0
Sequence and score formulas	None	P0
Progression/bypass/retry/idempotency	None	P0
Gamification deduplication	Mutable DB script only; production formulas differ	P0
Classes/enrollment/files/quizzes	Features and tests absent	P0
Score override/final/release	Features and tests absent	P0
Analytics/report reconciliation	None	P0
Accessibility/usability/performance/security	No results	P1

SECTION 39

Out-of-Scope / Unnecessary Features

No major scope-creep subsystem was found. There is no ERP, registrar, finance, HR, video-conferencing, multi-tenant, proctoring, 3D/AR/VR, official TESDA integration or real-device-control implementation.

Item	Audit treatment
Leaderboard	Within Objective 2 gamification, but it must not displace competency integrity work.
Bulk user import	Potentially within Admin account provisioning, but the current mock is not operational and should remain behind Admin RBAC.
Global notification/appearance settings	Secondary; retain only where they support the approved platform. Do not prioritize over P0 role/evaluation/class workflows.
Optional AI quiz generation	Explicit P1; do not begin before secure Instructor review/publish controls.

Detailed Gap Register

CAP-GAP-001

INCORRECT

P0

Web authentication is bypassed

System / severity	Web / Security / Critical
Evidence	src/components/auth/LoginForm.tsx; src/hooks/use-auth.tsx; protected-route.tsx
PRD / objective	INS-001, ADM-001, SHR-001, NFR-001; Objectives 4-5; Panel #2
Current behavior	Any email containing @ plus a non-empty password sets a localStorage flag; protected content always renders.
Expected behavior	Real session authentication and backend-enforced role/status checks.
Gap	No trustworthy web identity or protected route.
Recommended future action	Restore one approved identity architecture and enforce server-side session, status and role checks.

CAP-GAP-002

INCORRECT

P0

Instructor and Admin roles/workspaces are merged

System / severity	Web / RBAC / Critical
Evidence	src/types/index.ts; src/types/auth.ts; sidebar.tsx; topbar.tsx
PRD / objective	INS-001, ADM-001; Objectives 4-5; Panel #2
Current behavior	instructor_admin is the only staff role and sees teaching plus system administration.
Expected behavior	Separate roles, destinations, navigation, capabilities and backend policies.
Gap	Privilege separation is structurally impossible.
Recommended future action	Define distinct role/capability model and separate protected route trees.

CAP-GAP-003

INCORRECT

P0

Mobile and web do not share an authoritative platform

System / severity	Shared Architecture / Critical
Evidence	Mobile SupabaseConfig/services; web firebase.ts/firebase.service.ts
PRD / objective	SHR-001 to SHR-003; Objectives 3-5; Panel #2, #9
Current behavior	Mobile writes Supabase; web initializes Firebase and mostly displays constants.
Expected behavior	One reconciled identity/data/service model.
Gap	Cross-product user, class, content and score synchronization cannot occur.
Recommended future action	Confirm provider and data ownership, then document and implement a shared contract without inventing a rewrite.

CAP-GAP-004

MISSING

P0

No secure backend/API or repository authorization rules

System / severity	Backend / Security / Critical
Evidence	No src/app/api, middleware, server, schema, firestore.rules, storage.rules or Supabase policies
PRD / objective	SHR-002, NFR-001/002/010; Objectives 4-5; Panel #2, #7, #9-11
Current behavior	Clients directly perform database CRUD and send score values.
Expected behavior	Server-authoritative validation, object scope and critical mutation control.
Gap	Sensitive actions depend on browser/device trust.
Recommended future action	Introduce a minimal secure service boundary and provider-native policies with authorization tests.

CAP-GAP-005

MISSING

P0

Class, enrollment and assignment domain is absent

System / severity	Web / Mobile / Data / Critical
Evidence	No class/enrollment/assignment route, model or service
PRD / objective	MOB-002, INS-002/003/009/017, ADM-004; Objective 5; Panel #9
Current behavior	No owned classes or learner memberships exist.
Expected behavior	LMS-lite class lifecycle, enrollment history, assignments and mobile visibility.
Gap	Instructor scope, learner access and class analytics cannot be enforced.
Recommended future action	Implement the minimum class/enrollment/assignment model after identity/RBAC.

CAP-GAP-006

INCORRECT

P0

Account state, deactivation and deletion violate role policy

System / severity	Web / Mobile / Backend / High
Evidence	users/page.tsx; users/[id]/edit/page.tsx; mobile auth_service.dart
PRD / objective	INS-004, ADM-003, EVAL-009; Objectives 4-5; Panel #11
Current behavior	Combined UI toggles local state and deletes a Firestore document; mobile ignores active status at login.
Expected behavior	Instructor reasoned deactivation only; Admin guarded removal; historical integrity.
Gap	Deactivated accounts may enter; permanent deletion is improperly exposed and incomplete.
Recommended future action	Model lifecycle/reasons, enforce at authentication and reserve safeguarded removal to Admin.

CAP-GAP-007

MISSING

P0

Approved TESDA source/version and criteria are absent

System / severity	Content / Data / Critical
Evidence	No TESDA/rubric/criterion schema; competency/criteria routes redirect
PRD / objective	ADM-007, INS-007, EVAL-001/008; Objectives 3-4; Panel #1, #6
Current behavior	COC names and rules are embedded locally without source metadata.
Expected behavior	Approved immutable source version, COCs, criteria and weights.
Gap	No defensible TESDA traceability.
Recommended future action	Resolve PRD OD-01/OD-02 and encode only instructor-approved values.

CAP-GAP-008

INCORRECT

P0

Client hard-coded scoring replaces a criterion engine

System / severity	Mobile / Evaluation / Critical
Evidence	configuration/drag_drop/step_procedure templates; mission_result_model.dart
PRD / objective	MOB-009, EVAL-002/004; Objective 4; Panel #1, #6
Current behavior	UI templates compute unversioned weighted totals and hard-code 75 competence.
Expected behavior	Deterministic server-side criterion results from approved configuration.
Gap	Scores are not reproducible, authoritative or TESDA-grounded.
Recommended future action	Separate competency scoring from UI and gamification; implement versioned rule evaluation.

CAP-GAP-009

PARTIAL P0

Attempt action and criterion evidence is incomplete

System / severity	Mobile / Data / High
Evidence	result_service.dart; task_result_model.dart; step_procedure_mission_screen.dart
PRD / objective	MOB-008, EVAL-003/004; Objective 4; Panel #7
Current behavior	Some task outcomes are stored, but no complete chronological event stream or criterion match.
Expected behavior	Start/end, actions, order, errors, elapsed time, criteria and submission state.
Gap	Instructor cannot prove how a learner performed a required task.
Recommended future action	Define append-only attempt events and criterion evidence attached to the published version.

CAP-GAP-010

INCORRECT P0

Duplicate and partial mission completion is possible

System / severity	Mobile / Backend / Critical
Evidence	enhanced_templates _saveResultsToDatabase; ResultScreen._saveResult; MissionCompletionService
PRD / objective	MOB-013, SHR-006, NFR-002/005; Objectives 2, 4, 5; Panel #6, #7
Current behavior	Enhanced flow persists before navigation, then ResultScreen persists again; multi-step writes are non-atomic.
Expected behavior	One idempotent final submission and transactional/equivalent updates.
Gap	Duplicate result/task rows, attempts and inconsistent rewards/reports.
Recommended future action	Create a unique submission identity and authoritative atomic completion orchestration.

CAP-GAP-011

MISSING P0

Provisional, final and released result states are absent

System / severity	Mobile / Web / Data / Critical
Evidence	No final_score/release/review state; result_screen.dart shows immediate generic result
PRD / objective	MOB-010, INS-014, EVAL-006; Objective 4; Panel #7
Current behavior	Client result is treated as passed/competent immediately.
Expected behavior	Provisional result, Instructor decision, immutable final revision and release state.
Gap	Learner cannot distinguish automated from authoritative outcome.
Recommended future action	Add assessment review/final/release lifecycle after criterion evaluation.

CAP-GAP-012

MISSING P0

Instructor score review and adjustment is absent

System / severity	Web / Backend / Critical
Evidence	No assessment review route/model/service
PRD / objective	INS-012/013/014, EVAL-005/006; Objective 4; Panel #7
Current behavior	Only dummy score tables exist.
Expected behavior	Evidence review, confirm/edit, reason, remarks, affected criteria, finalization and history.
Gap	Panel recommendation and core automated-evaluation governance are unmet.
Recommended future action	Implement Instructor-scoped review using immutable revision records.

CAP-GAP-013**MISSING****P0****Audited COC bypass is absent**

System / severity	Web / Mobile / Data / High
Evidence	No bypass model/service/UI; mobile 50% rule
PRD / objective	INS-010, EVAL-007; Objective 3; Panel #3
Current behavior	No authorized bypass; progression is local automatic logic.
Expected behavior	Scope/reason/date/instructor audit; unlock only.
Gap	Required exception workflow and states do not exist.
Recommended future action	Add access-only bypass after class/permission model; prohibit auto-pass/reward.

CAP-GAP-014**PARTIAL****P0****Instructor content and quiz publishing is non-operational**

System / severity	Web / Data / High
Evidence	modules hard-coded; edit TODO; scenario/criteria routes redirect; no quiz routes
PRD / objective	INS-005/007; Objectives 1, 3; Panel #1, #8
Current behavior	Content is displayed and an edit form toasts success without persistence.
Expected behavior	Draft/preview/validate/publish/archive versioned content and quizzes.
Gap	Mobile content cannot be instructor-governed or traced.
Recommended future action	Build minimal versioned content workflow mapped to approved criteria.

CAP-GAP-015**MISSING****P0****Learning-resource file management is absent**

System / severity	Web / Storage / High
Evidence	No resource upload/model/storage code
PRD / objective	INS-006, SHR-005; Objectives 1, 3; Panel #10, #12
Current behavior	No PDF/image/video resource service.
Expected behavior	Validated scoped upload/download/stream, metadata, version and audit.
Gap	Panel file and video requirements cannot be met.
Recommended future action	Implement after storage provider and class/content authorization are fixed.

CAP-GAP-016**INCORRECT****P0****Instructor analytics and reports are fabricated**

System / severity	Web / Reporting / High
Evidence	dashboard/analytics/progress/reports DUMMY_* and Math.random
PRD / objective	INS-011/015/016, ADM-012; Objective 5; Panel #10
Current behavior	Charts and CSV use constants/random values.
Expected behavior	Aggregates and exports reconciled to stored attempts/final scores and scope.
Gap	Objective 5 cannot be demonstrated with real learners.
Recommended future action	Replace only after shared data/evaluation truth is established.

CAP-GAP-017

INCORRECT P0

Audit logging is incomplete and unattributed

System / severity	Web / Shared / High
Evidence	firebase.service.ts::logAction; logs/page.tsx
PRD / objective	ADM-009, SHR-009, NFR-003; Objectives 4-5; Panel #3, #7, #11
Current behavior	Some UI actions log display text; auth.currentUser is null under bypass; critical changes omitted.
Expected behavior	Immutable actor ID, role, time, action, target, old/new, reason/outcome.
Gap	Critical decisions cannot be reconstructed.
Recommended future action	Create centralized audit writes within authoritative mutation transactions.

CAP-GAP-018

MISSING P0

Content/rubric/score history is not versioned

System / severity	Shared Data / High
Evidence	No version/revision entities or fields
PRD / objective	SHR-004/007, EVAL-006/008, NFR-013; Objectives 3-5; Panel #1, #7
Current behavior	Hard-coded content can change; result lacks source/rubric version.
Expected behavior	Attempts preserve immutable published versions and score revisions.
Gap	Historical reports cannot be reproduced.
Recommended future action	Introduce version references before publishing new assessments.

CAP-GAP-019

MISSING P0

Learner assignments, retry limits and windows are absent

System / severity	Mobile / Web / High
Evidence	No class assignment/retry/availability model
PRD / objective	MOB-002/012, INS-009; Objectives 1, 5; Panel #9
Current behavior	Local mission unlocks control access.
Expected behavior	Instructor-configured class availability and retry policy preserving history.
Gap	Learner access cannot match class policy.
Recommended future action	Add after classes and versioned content.

CAP-GAP-020

INCORRECT P0

Simulation panel UI corrections are incomplete

System / severity	Mobile UI / High
Evidence	main.dart portrait lock; no InteractiveViewer; visible drop zones; no video dependency
PRD / objective	MOB-004/006/007/014, UI-001/002/004/005; Objective 1; Panel #4, #5, #12, #14
Current behavior	Text-only portrait UI with permanent targets and prominent imagery.
Expected behavior	Zoom/full-screen, hidden assessment targets, optional video and balanced hierarchy.
Gap	Demonstration/accessibility and panel acceptance fail.
Recommended future action	Apply after assessment/practice states are defined so hints are mode-aware.

CAP-GAP-021

UNVERIFIED

P0

External database schema and policies are unavailable

System / severity	Data / Security / High
Evidence	16 Supabase names referenced; no schema/migration/RLS/rules files
PRD / objective	SHR-001 to SHR-009; Objectives 2-5; Panel #1, #2, #6-11
Current behavior	Static clients assume records/policies.
Expected behavior	Documented schema, keys, constraints, policies, functions and backup ownership.
Gap	Persistence/security cannot be verified.
Recommended future action	Export/read-only document the actual provider schema and policies before coding.

CAP-GAP-022

MISSING

P0

Objective-critical automated tests are absent

System / severity	QA / High
Evidence	No web tests; mobile widget smoke; mutable db_test.dart
PRD / objective	NFR-014; Definition of Done; All objectives; Panel All
Current behavior	No role, scoring, evidence, idempotency, class, score or analytics tests.
Expected behavior	Repeatable unit/integration/security/usability/performance evidence.
Gap	No objective can meet Definition of Done.
Recommended future action	Build tests alongside each P0 domain; isolate external services/emulators.

CAP-GAP-023

INCORRECT

P0

Incorrect RJ45 question answer key

System / severity	Mobile Content / High
Evidence	mission_content_data.dart lines 522-524
PRD / objective	MOB-005, EVAL-002; Objectives 1, 4; Panel #1
Current behavior	Option 'RJ45 Connector' does not equal correctAnswer 'RJ45'.
Expected behavior	Every expected answer is selectable and criterion-tested.
Gap	A learner cannot answer correctly under exact matching.
Recommended future action	Validate content keys against options through automated fixtures after rubric confirmation.

CAP-GAP-024

MISSING

P0

Supplementary/non-certification disclosure is absent

System / severity	Mobile / Web / High
Evidence	No disclaimer string found
PRD / objective	EVAL-010; Scope 14.1; Objectives 3-4; Panel #1
Current behavior	CSS NC II labels appear without the required limitation statement.
Expected behavior	Prominent statement that results are supplementary, not official TESDA certification.
Gap	Scope/ethics acceptance condition unmet.
Recommended future action	Add approved wording to relevant learner/result/report surfaces later.

CAP-GAP-025

INCORRECT P0

Web environment contract does not match implementation

System / severity	Web Configuration / High
Evidence	<code>.env.example/.env.local</code> names vs <code>src/lib/firebase.ts</code>
PRD / objective	SHR-010, NFR-009; Objectives 4-5; Panel #2
Current behavior	Supabase names supplied; Firebase client expects different variables and falls back to mock-key.
Expected behavior	Confirmed documented shared provider configuration.
Gap	Real web data/auth initialization is not reproducible.
Recommended future action	Resolve provider decision and align documented variables without exposing secrets.

CAP-GAP-026

MISSING P1

Optional AI quiz drafting is absent

System / severity	Web / Backend / Medium
Evidence	No quiz/AI SDK/server route
PRD / objective	INS-008, ADM-014; Objective 3 support; Panel #8
Current behavior	No AI feature.
Expected behavior	Optional draft-only generation with provenance and mandatory Instructor review.
Gap	Panel optional recommendation not implemented.
Recommended future action	Defer until all P0 identity, content, rubric and publishing controls pass.

CAP-GAP-027

MISSING P1

Backup/recovery and administrative governance are absent

System / severity	Admin / Backend / High
Evidence	No backup/restore/file-governance services
PRD / objective	ADM-006/010/011, SHR-008, NFR-011; Objectives 4-5 continuity; Panel #10, #11
Current behavior	Only UI settings and dummy CSV report export.
Expected behavior	Essential-data export, tested restore, file limits/types and audited governance.
Gap	Defense continuity and recovery cannot be demonstrated.
Recommended future action	Implement after the authoritative schema and retention policy are confirmed.

CAP-GAP-028

INCORRECT P1

Gamification has competing local and remote truth

System / severity	Mobile / Medium
Evidence	<code>LocalGamificationService</code> ; <code>GamificationService</code> ; <code>ProfileService</code>
PRD / objective	MOB-011, SHR-003/006; Objective 2; Panel #6
Current behavior	<code>SharedPreferences</code> and Supabase separately calculate/update XP/levels; reconciliation uses maximum values.
Expected behavior	One authoritative award event and deterministic projection.
Gap	Drift, duplicate rewards and cross-screen inconsistency remain possible.
Recommended future action	Assign one source of truth and make local state a cache/projection.

CAP-GAP-029

PARTIAL P1

Accessibility evidence is insufficient

System / severity	Mobile / Web UI / Medium
Evidence	0 Flutter Semantics/semanticLabel/Tooltip; sparse ARIA; no zoom
PRD / objective	UI-001/007, NFR-007; Objective 1 usability; Panel #4, #14
Current behavior	Responsive layouts and labels exist but assistive semantics are sparse.
Expected behavior	Scalable/zoomable, keyboard/screen-reader and touch-accessible workflows.
Gap	Accessibility acceptance cannot be demonstrated.
Recommended future action	Create a focused audit and device/keyboard/screen-reader test set.

CAP-GAP-030

UNVERIFIED P1

Mobile release/build configuration is not defense-ready

System / severity	Mobile Platform / Medium
Evidence	android/app/build.gradle and build.gradle.kts; debug signing; com.example.bytequest
PRD / objective	NFR-008/009; Deployment support; Panel N/A
Current behavior	Two differing Gradle files coexist and release uses debug identity/signing.
Expected behavior	One confirmed target configuration and repeatable release build.
Gap	Production package identity/compatibility unverified.
Recommended future action	Resolve only after approved deployment targets (OD-04) are confirmed.

CAP-GAP-031

PARTIAL P2

Asset and documentation contracts drift from code

System / severity	Mobile / Documentation / Low
Evidence	pubspec missing directories; web/mobile summary docs claim completed behavior contradicted by source
PRD / objective	NFR-009; Definition of Done documentation; All; Panel #14
Current behavior	Declared asset directories are absent; prose reports overstate real auth/integration.
Expected behavior	Build-safe manifests and documentation matching current behavior.
Gap	Handoff and defense claims may be misleading.
Recommended future action	Reconcile documentation and manifests after implementation stabilizes.

CAP-GAP-032

PARTIAL P2

Deeper analytics and UI polish remain enhancements

System / severity	All UI / Reporting / Low
Evidence	Placeholder chart on mobile; dashboard mock trends; encoding artifacts
PRD / objective	P2/P1 enhancements after baseline; Objective 5; Panel #14
Current behavior	Visual shells exist with limited trustworthy insight and minor polish defects.
Expected behavior	Evidence-linked intervention analytics and final visual QA after core workflows.
Gap	Polish cannot compensate for missing source data.
Recommended future action	Defer until integrated data and tests are complete.

P0 - Critical Development Gaps

P0 items block study objectives, mandatory PRD workflows, security or data integrity. Complete them in dependency order, not UI order.

Gap ID	Title	System	Severity	Dependency / future action
CAP-GAP-001	Web authentication is bypassed	Web / Security	Critical	Restore one approved identity architecture and enforce server-side session, status and role checks.
CAP-GAP-002	Instructor and Admin roles/workspaces are merged	Web / RBAC	Critical	Define distinct role/capability model and separate protected route trees.
CAP-GAP-003	Mobile and web do not share an authoritative platform	Shared Architecture	Critical	Confirm provider and data ownership, then document and implement a shared contract without inventing a rewrite.
CAP-GAP-004	No secure backend/API or repository authorization rules	Backend / Security	Critical	Introduce a minimal secure service boundary and provider-native policies with authorization tests.
CAP-GAP-005	Class, enrollment and assignment domain is absent	Web / Mobile / Data	Critical	Implement the minimum class/enrollment/assignment model after identity/RBAC.
CAP-GAP-006	Account state, deactivation and deletion violate role policy	Web / Mobile / Backend	High	Model lifecycle/reasons, enforce at authentication and reserve safeguarded removal to Admin.
CAP-GAP-007	Approved TESDA source/version and criteria are absent	Content / Data	Critical	Resolve PRD OD-01/OD-02 and encode only instructor-approved values.
CAP-GAP-008	Client hard-coded scoring replaces a criterion engine	Mobile / Evaluation	Critical	Separate competency scoring from UI and gamification; implement versioned rule evaluation.
CAP-GAP-009	Attempt action and criterion evidence is incomplete	Mobile / Data	High	Define append-only attempt events and criterion evidence attached to the published version.
CAP-GAP-010	Duplicate and partial mission completion is possible	Mobile / Backend	Critical	Create a unique submission identity and authoritative atomic completion orchestration.
CAP-GAP-011	Provisional, final and released result states are absent	Mobile / Web / Data	Critical	Add assessment review/final/release lifecycle after criterion evaluation.
CAP-GAP-012	Instructor score review and adjustment is absent	Web / Backend	Critical	Implement Instructor-scoped review using immutable revision records.
CAP-GAP-013	Audited COC bypass is absent	Web / Mobile / Data	High	Add access-only bypass after class/permission model; prohibit auto-pass/reward.
CAP-GAP-014	Instructor content and quiz publishing is non-operational	Web / Data	High	Build minimal versioned content workflow mapped to approved criteria.
CAP-GAP-015	Learning-resource file management is absent	Web / Storage	High	Implement after storage provider and class/content authorization are fixed.
CAP-GAP-016	Instructor analytics and reports are fabricated	Web / Reporting	High	Replace only after shared data/evaluation truth is established.
CAP-GAP-017	Audit logging is incomplete and unattributed	Web / Shared	High	Create centralized audit writes within authoritative mutation transactions.
CAP-GAP-018	Content/rubric/score history is not versioned	Shared Data	High	Introduce version references before publishing new assessments.

Gap ID	Title	System	Severity	Dependency / future action
CAP-GAP-019	Learner assignments, retry limits and windows are absent	Mobile / Web	High	Add after classes and versioned content.
CAP-GAP-020	Simulation panel UI corrections are incomplete	Mobile UI	High	Apply after assessment/practice states are defined so hints are mode-aware.
CAP-GAP-021	External database schema and policies are unavailable	Data / Security	High	Export/read-only document the actual provider schema and policies before coding.
CAP-GAP-022	Objective-critical automated tests are absent	QA	High	Build tests alongside each P0 domain; isolate external services/emulators.
CAP-GAP-023	Incorrect RJ45 question answer key	Mobile Content	High	Validate content keys against options through automated fixtures after rubric confirmation.
CAP-GAP-024	Supplementary/non-certification disclosure is absent	Mobile / Web	High	Add approved wording to relevant learner/result/report surfaces later.
CAP-GAP-025	Web environment contract does not match implementation	Web Configuration	High	Resolve provider decision and align documented variables without exposing secrets.

SECTION 42

P1 - Important Development Gaps

P1 work completes important resilience, accessibility, governance and optional panel expectations after the P0 platform is stable.

Gap ID	Title	System	Severity	Dependency / future action
CAP-GAP-026	Optional AI quiz drafting is absent	Web / Backend	Medium	Defer until all P0 identity, content, rubric and publishing controls pass.
CAP-GAP-027	Backup/recovery and administrative governance are absent	Admin / Backend	High	Implement after the authoritative schema and retention policy are confirmed.
CAP-GAP-028	Gamification has competing local and remote truth	Mobile	Medium	Assign one source of truth and make local state a cache/projection.
CAP-GAP-029	Accessibility evidence is insufficient	Mobile / Web UI	Medium	Create a focused audit and device/keyboard/screen-reader test set.
CAP-GAP-030	Mobile release/build configuration is not defense-ready	Mobile Platform	Medium	Resolve only after approved deployment targets (OD-04) are confirmed.

SECTION 43

P2 - Enhancements

P2 work is polish and optimization. It must not delay the objective-critical backlog.

Gap ID	Title	System	Severity	Dependency / future action
CAP-GAP-031	Asset and documentation contracts drift from code	Mobile / Documentation	Low	Reconcile documentation and manifests after implementation stabilizes.
CAP-GAP-032	Deeper analytics and UI polish remain enhancements	All UI / Reporting	Low	Defer until integrated data and tests are complete.

Recommended Development Sequence

PHASE 1

PHASE 1 - P0 Objective-Critical Gaps

Confirm provider/current production schema; restore real authentication; split roles/capabilities; enforce active account state; add secure server/policy boundary; define common IDs and idempotent submission.

Dependency: prerequisite for every protected class, score, file, audit and report workflow.

PHASE 2

PHASE 2 - TESDA / Evaluation Alignment

Resolve OD-01/OD-02; register approved source version; model COCs, published module versions, criteria and weights; implement deterministic criterion evaluation; separate competency score from rewards; preserve attempt version/evidence.

Dependency: identity/data foundation from Phase 1; values must be instructor-approved.

PHASE 3

PHASE 3 - Instructor Workflow

Classes, enrollment, assignments, retry windows, content/quiz publish, files, COC bypass, real monitoring, evidence review, score adjustment, finalization and release.

Dependency: Phases 1-2 and immutable audit service.

PHASE 4

PHASE 4 - Admin / RBAC

Dedicated Admin workspace; account lifecycle; instructor scope; TESDA version activation; settings/file governance; system audit; guarded deletion and emergency score privilege.

Dependency: capability matrix and class/content/evaluation records.

PHASE 5

PHASE 5 - Analytics and Reporting

Replace constants with scoped queries/aggregates; reconcile learner, class, module, competency and final score totals; implement approved exports/snapshots.

Dependency: trustworthy attempt/final-score records from Phases 2-3.

PHASE 6

PHASE 6 - UX / Panel Improvements

Zoom/full-screen, hidden assessment targets, practice hints, video scenario delivery, explicit state labels, balanced imagery, accessibility and encoding polish.

Dependency: practice/assessment and release-state policies defined.



PHASE 7

PHASE 7 - Testing and Final Validation

Unit/property tests for rules; integration tests for mobile-web data; authorization matrix; idempotency/concurrency; content fixtures; accessibility, performance, recovery, privacy and adviser/TESDA validation.

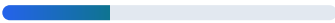
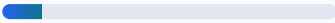
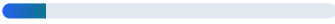
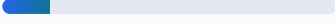
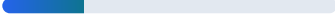
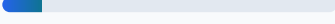
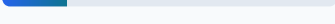
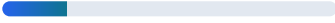
Dependency: tests begin in every phase; this phase completes end-to-end validation and defense evidence.

P1 AI gate: evaluate AI quiz drafting only after Phase 7 confirms secure draft/review/publish controls. It must never grade competency or auto-publish.

Readiness / Completion Assessment

Scoring Method

Each explicit requirement receives a status factor: COMPLETE 1.00, PARTIAL 0.50, INCORRECT 0.20, UNVERIFIED 0.25, MISSING 0.00; OUT OF SCOPE is excluded. Priority weights are P0 = 5, P1 = 3 and P2 = 1. A requirement contributes status factor x priority weight. Percentages are rounded to the nearest whole percent.

Area	Weighted basis	Readiness	Interpretation
Mobile Application	22.0 / 69 weighted points	32% 	20 missions and five interactions provide meaningful credit; role/class/TESDA/evaluation/finality/ident potency gaps dominate.
Instructor Dashboard	10.0 / 83	12% 	Presentation screens exist, but all core class/content/evidence/score workflows are absent or mocked.
Admin Dashboard	8.5 / 68	13% 	Some account/settings/log layouts exist; role separation and governed backend behavior do not.
Backend / API	Mapped 16-control checklist	14% 	Direct client service code receives partial credit; no server API, rules artifact, authoritative validation or transactions.
Database / Data Model	Mapped 24-domain checklist	24% 	Core mobile progress/gamification names exist, but schema is unverified and class/rubric/final/audit/file domains are absent.
TESDA Alignment	Mapped source/criteria/version/evaluation controls	12% 	Topical COC coverage earns partial credit; approved traceability and rules do not exist.
Objective Compliance	165 / 500 objective-rubric points	33% 	Objective audit scores: 55, 55, 30, 10 and 15 percent.
Panel Recommendations	12.0 / 64 priority-weighted points	19% 	Interaction diversity complete; text scenarios and visual balance partial; core role/evaluation/class recommendations open.
Overall System	79.25 / 414 PRD-weighted points	19% 	Aggregate of all 88 explicit PRD IDs; not an average of subsystem percentages.

How to use the estimates

These percentages measure demonstrable PRD compliance, not source-code volume, visual completion or development effort already spent. Mock screens receive UI credit but no persistence/backend credit. External services without repository definitions receive UNVERIFIED rather than COMPLETE. A later runtime audit may adjust the values once the actual schema, RLS/rules, deployed providers and test results are supplied.

Final Conclusions

ByteQuest is presently a visually convincing and technically meaningful prototype, not a PRD-compliant assessment platform. Its strongest asset is the learner-side mission catalog; its largest deficit is the absence of a common secured platform for versioned competency evidence, Instructor review and role-governed administration.

Direct Answers to the Audit Questions

1. **Currently implemented:** 20 mobile missions, five interaction types, text scenarios, mobile progress/gamification, dashboard presentation shells and limited direct client persistence.
2. **Fully complete:** interaction-type variety and several scope boundaries; no complete integrated study objective.
3. **Partially implemented:** learner profile/modules/progress, text scenarios, gamification, result storage and visual monitoring/report layouts.
4. **Missing:** shared backend, classes/enrollment, TESDA versions/criteria, file/quiz workflows, bypass, instructor finalization, release, backup and objective-level tests.
5. **Incorrect:** web auth/RBAC, merged roles, client-trusted scores, hard-coded evaluation, visible drag targets, dummy analytics, user deletion and duplicate completion.
6. **Unverified:** external Supabase schema/RLS/data, Firebase project/rules, production builds, deployed performance and end-to-end synchronization.
7. **Objective blockers:** lack of approved TESDA traceability, criterion evidence, secure server evaluation, instructor workflow and real analytics.
8. **Panel status:** interaction diversity is satisfied; text scenario support and visual balance are partial; the remaining core recommendations require work.
9. **Priority:** establish identity/RBAC/shared data, TESDA/rubric model and idempotent evaluation first; then Instructor/Admin workflows and analytics; apply UI panel corrections and QA before P1 AI.
10. **Distance to acceptance:** the severity-weighted PRD readiness is 19%; this measures demonstrated compliance, not raw code volume.

Final no-guess statement. TESDA value/rule requires validation against the approved TESDA CSS NC II booklet.

Read-Only Verification Declaration

The full PRD, Flutter source, Next.js source, referenced persistence integrations, role boundaries, security surfaces and tests were inspected. All five study objectives and all 14 panel recommendations were evaluated. Missing capability was not upgraded based on filenames or planning documents. No application source, dependency, configuration, environment, database, migration, UI, commit or remote repository state was modified.